

# NANENG 222 Final Project

## Audio Amplifier

|                        |           |
|------------------------|-----------|
| Mohamed Khaled Mohamed | 202401477 |
| Mohab Mahmoud          | 202402208 |
| Youssef Ahmed          | 202401212 |
| Marwan Darrag          | 202300661 |
| Hassan Emad            | 202401299 |

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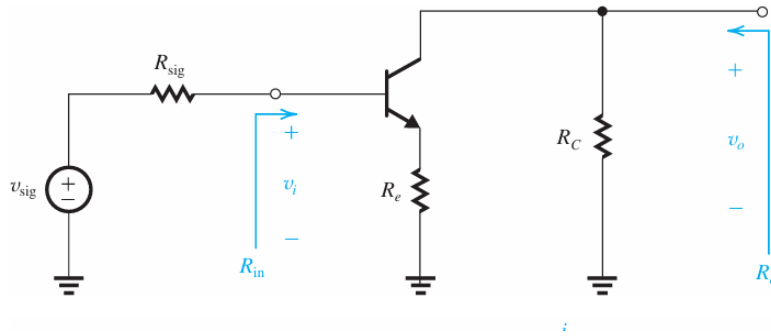
## Theory

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### Common Emitter Stage with Degeneration

#### *Gain Analysis*

7.3 Basic C



$$i_e = \frac{v_{in}}{\alpha r_m + R_E} \quad (1)$$

$$v_o = -\alpha i_e R_C = -\frac{\alpha R_C}{\alpha r_m + R_E} v_{in} \quad (2)$$

$$A_v = \frac{-\alpha R_C}{\alpha r_m + R_E} \quad (3)$$

Side note:  $\alpha \simeq 1$ ,  $r_m \sim 25\text{--}100 \, \Omega$

$$A_v = \frac{\text{total resistance between Collector and AC ground}}{\text{total resistance between Emitter and AC ground}} \quad (4)$$

### Approximations

$$A_v \simeq -\frac{R_C}{R_E} \rightarrow \text{no dependence on PVT variations} \quad (5)$$

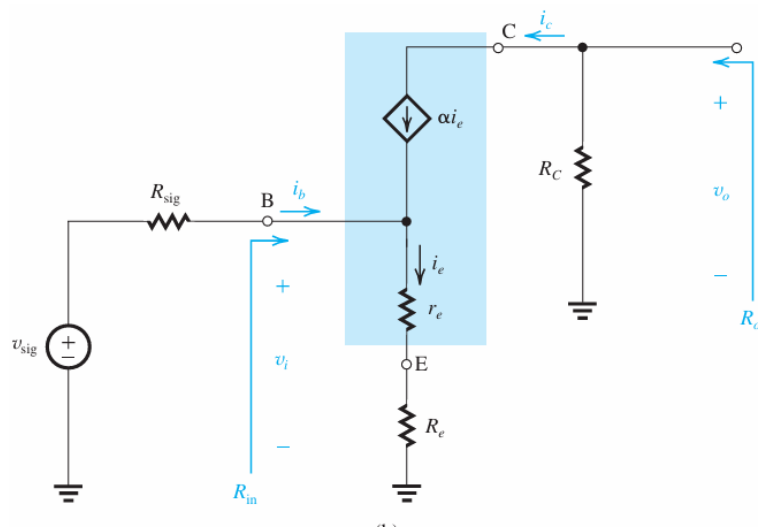
- **P** →  $R_C/R_E$  scale by the same ratio
- **V** →  $r_m$  is neglected, which is biasing
- **T** → we neglect  $r_m$

as  $R_E \gg r_m$

### Insights

$$A_v = -\frac{\alpha R_C g_m}{\alpha + R_E g_m} = -\frac{R_C g_m}{1 + \frac{R_E}{\alpha} g_m} \quad (6)$$

*Note on the denominator:* → lower gain because  $R_E$ .



### Input and Output Impedance

#### Input Impedance

We will do it without including the early effect (it has negligible effect):

$$i_x = (1 - \alpha) i_e \quad (7)$$

$$i_x = (1 - \alpha) \frac{v_x}{\alpha r_m + R_E} \quad (8)$$

$$R_{in} = \frac{\alpha r_m + R_E}{1 - \alpha} = \frac{\alpha}{1 - \alpha} r_m + \frac{R_E}{1 - \alpha} \quad (9)$$

$$= \beta r_m + \frac{R_E}{1 - \alpha} = r_\pi + \frac{R_E}{1 - \alpha} \quad (10)$$

$$R_{in} \simeq r_\pi + R_E(\beta + 1) \quad (11)$$

- $R_{in} \rightarrow k\Omega$
- $R_E(\beta + 1)$  term:  $\sim 100\ k\Omega \rightarrow$  degeneration

$$R_{in} \simeq r_\pi + R_E \beta \quad (12)$$

*Output Impedance*

$$i_x = \frac{v_x}{R_C} + \alpha i_e + \frac{v_x}{r_o} \quad (13)$$

*Annotation:*  $\alpha r_m \parallel R_E + r_o \simeq r_o$

$$\frac{v_x}{R_o} = \frac{v_x}{R_C} + \frac{v_x}{r_o} + \alpha \frac{v_x}{r_o} \frac{R_E}{R_E + \alpha r_m} \quad (14)$$

$$\frac{1}{R_o} = \frac{1}{R_C} + \frac{1}{r_o} + \frac{\alpha}{r_o} \frac{R_E}{R_E + \alpha r_m} \quad (15)$$

$$R_o = r_o \frac{R_E + \alpha r_m}{\alpha r_m + R_E(1 - \alpha)} \quad (16)$$

$$R_o = r_o \frac{\frac{R_E}{\alpha} + r_m}{\frac{R_E}{\beta} + r_m} \quad (17)$$

### Output Impedance Special Cases & Gain Limits

1)  $R_E < r_m$

$$R_o \simeq r_o \quad (18)$$

2)  $r_m < R_E \ll \beta r_m$

$$R_o \simeq r_o [1 + g_m R_E] \uparrow\uparrow \text{ and non-limited} \quad (19)$$

3)  $R_E \gg \beta r_m$

$$R_o \simeq r_o \frac{\beta}{\alpha} \simeq r_o \beta \quad (20)$$

### Side Notes on Gain Limits:

- Very high — this limits the gain
- That is very important
- Mean gain is limited
- In MOSFET:  $\alpha = 1, \beta = \infty$

$$R_o = r_o \frac{R_E + \alpha r_m}{\frac{R_E}{\beta} + r_m} = r_o [1 + g_m R_E] \quad (21)$$

Non-limited!

### Gain Calculations

Gain as we think:

$$A_v = \frac{-\alpha(R_C \parallel R_o)}{\alpha r_m + R_E} \quad (22)$$

Maximum gain:

$$A_{v_{\max}} = \frac{-\alpha R_C}{R_E} = -\frac{R_C}{R_E} \quad (23)$$

### Summary: Input Impedance & Voltage Gain

$$R_{in} = r_\pi + (\beta + 1)R_E \quad (24)$$

$$A_v = -\frac{\alpha(R_C \parallel R_o)}{\alpha r_m + R_E} \simeq -\frac{R_C}{r_m + R_E} \quad (25)$$

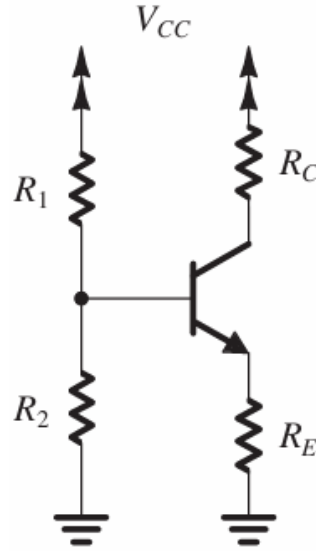
| Equation                                                                                  | Note                   | Approximation Used                                                      |
|-------------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------|
| $R_o = r_o \frac{\frac{R_E}{\alpha} + r_m}{\frac{R_E}{\beta} + r_m} \parallel R_C$        | Standard Derivation    | Using $r_o$ is large compared to $r_m$ and $R_E$                        |
| $= r_o \left[ \frac{R_E g_m + 1}{\frac{\alpha}{\beta} R_E g_m + 1} \right] \parallel R_C$ | Algebraic              | None (substituted                                                       |
| $\simeq r_o [1 + g_m R_E] \parallel R_C$                                                  | Rearrangement          | $r_m = \frac{1}{g_m}$ )                                                 |
| $= [(1 + g_m(r_\pi \parallel R_E)) r_o + (r_\pi \parallel R_E)] \parallel R_C$            | Standard Approximation | Assumes large $\beta$ ;<br>$\frac{\alpha}{\beta} R_E g_m \rightarrow 0$ |
| $= [1 + g_m r_o](r_\pi \parallel R_E) + r_o$                                              | Exact                  | None (alternative using $r_\pi$ )                                       |
|                                                                                           | “Exact” (Expanded)     | None (expansion looking into collector before $R_C$ )                   |

### Biasing Technique

In real practical circuits, we must bias all things by one voltage source. We want stable bias that does not depend on PVT variations—this is a big branch in analog.

Simply: how to get smaller voltages from a voltage source? Yeah, a voltage divider. Right.

$$V_B \simeq V_{CC} \cdot \frac{R_2}{R_1} \quad (26)$$



**Questions:**

**1) How this affects the small-signal model:**

A partial small-signal diagram looking into the base terminal shows the input impedance  $R_{in}$  and the parallel combination  $R_1 \parallel R_2$  to ground.

$$R'_{in} = R_1 \parallel R_2 \parallel R_{in} \quad (27)$$

$$R_{in} = r_{\pi} + \frac{R_E}{1 - \alpha} = r_{\pi} + \beta \frac{R_E}{\alpha} \simeq r_{\pi} + \beta R_E \quad (28)$$

So  $R_1, R_2$  must be large to avoid degrading  $R'_{in}$ .

**$R_o$ :**

A small-signal diagram looking into the collector terminal to find  $R_o$ , with  $R_1 \parallel R_2$  connected between the base and ground.

If  $V_A \simeq \infty$ , replace every  $\alpha r_m + R_E$  by  $\alpha r_m + R_E \parallel R_1 \parallel R_2$ .

$$R_o = R_C \quad (29)$$

It has small effect on  $R_o$ .

**Gain:**

A small-signal diagram showing an input voltage  $v_{in}$  at the base, with the base biased by  $R_1 \parallel R_2$  to ground.

Use the Reflection Rule.  $R_E + \alpha r_m$  is replaced by  $\alpha r_m + R_E \parallel R_1 \parallel R_2$ :

$$\text{Gain} = \frac{-\alpha R_C}{(\alpha r_m + R_E) \parallel R_1 \parallel R_2} \quad (30)$$

Gain increases.

### Design Approach for Biasing:

$$I_C, V_{CE} \rightarrow (\text{specs or chosen}) \xrightarrow{1/\beta} I_B \quad (31)$$

#### Step 1:

$$\frac{V_{CC}}{R_1 + R_2} \gg 10 I_B \simeq 10 I_B \quad (32)$$

You have a certain  $V_{B_{th}}$  or  $\simeq 0.8$  V.

#### Step 2:

$$\frac{R_2}{R_1 + R_2} V_{CC} \simeq V_{B_{th}} \simeq 0.8 \text{ V} \quad (33)$$

Keep  $\frac{R_2}{R_1 + R_2}$  constant; increasing  $R_1 + R_2$  is better.

### Coupling Stages

#### *Interfacing Circuit Stages*

How to interface circuit stages without destroying the biasing:

- a) Design all the circuit blocks with the same biasing, which is pretty hard. (DC Coupling)
- b) AC Coupling: move DC, prevent AC – a coupling capacitor blocks DC and passes AC.

A simple block diagram shows “Block 1” connected to “Block 2” via a series capacitor  $C$ .

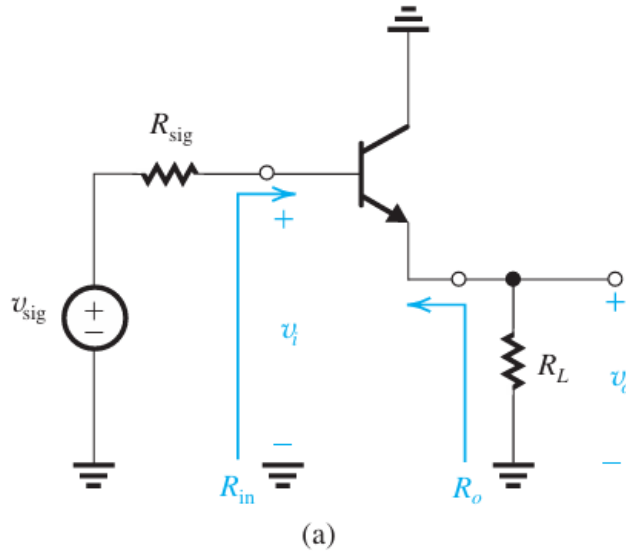
#### How to choose $C$ :

Make  $C$  large enough so it is a short circuit compared to the impedance environment of the circuit.

Side note: The tradeoff is speed.

## Emitter Follower

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1)  $R_{in}$  (same as Common Emitter)

$$R_{in} = r_{\pi} + (\beta + 1)R_E \quad (34)$$

2) Output Impedance:

$$R_o = R_E \parallel (\text{resistance looking into Emitter}) = R_E \parallel \alpha r_m \quad (35)$$

3) Gain:

A small-signal voltage-divider circuit:  $v_{in}$  goes through series resistance  $\alpha r_m$  to the output node  $v_o$ , which is connected to ground via  $R_E$ .

$$A_v = \frac{R_E}{R_E + \alpha r_m} \simeq 1 \quad (36)$$

Using the same biasing technique as the last stage:

It changes mainly  $R_{in}$ :

$$R_{in} = (r_{\pi} + (\beta + 1)R_E) \parallel R_1 \parallel R_2 \quad (37)$$

So we want  $R_1, R_2 \gg r_{\pi} + (\beta + 1)R_E$  to avoid degrading  $R_{in}$ .

## Design Approach

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**Goal:** Gain  $A_v = 4$  with tunable gain.

**Choose Stable Biasing:**

$$A_v \simeq -\frac{\alpha R_C}{\alpha r_m + R_E} \simeq -\frac{R_C}{r_m + R_E} \quad (38)$$

We want  $R_E \gg r_m$  where  $r_m = V_T/I_C$ . Since  $V_T$  is in the mV range we want  $I_C$  in the mA range:

$$I_C \simeq \frac{V_A}{R_C + R_E} \quad (39)$$

With  $V_A$  in the range of 100 V, we want  $R_C + R_E$  in the  $\sim k\Omega$  range.

Then:

$$R_{in} = [r_\pi + (\beta + 1)R_E] \parallel R_1 \parallel R_2 \simeq (\beta + 1)R_E \parallel R_1 \parallel R_2 \quad (40)$$

$R_1, R_2$  in the  $k\Omega$  range. Choose  $R_1, R_2$  to make stable bias:

$$\frac{V_{CC} R_2}{R_1 + R_2} \gg 10 I_B \sim 10 \mu A \quad (41)$$

$$V_B \simeq 0.8 \text{ V to } 1.5 \text{ V} \quad (42)$$

Selected component values:

$$R_2 = 2 \text{ k}\Omega, \quad R_1 = 10 \text{ k}\Omega, \quad R_E = 1 \text{ k}\Omega, \quad R_C = 4 \text{ k}\Omega$$

Now  $A_v = -4$  without dependence on P, V, T variations.

**Making the gain variable:** Split  $R_E$  into  $R_{E1}$  and  $R_{E2}$  in series, but place a capacitor in parallel with one of them. Keep  $R_{E1} + R_{E2} = 1 \text{ k}\Omega$  so the bias does not change:

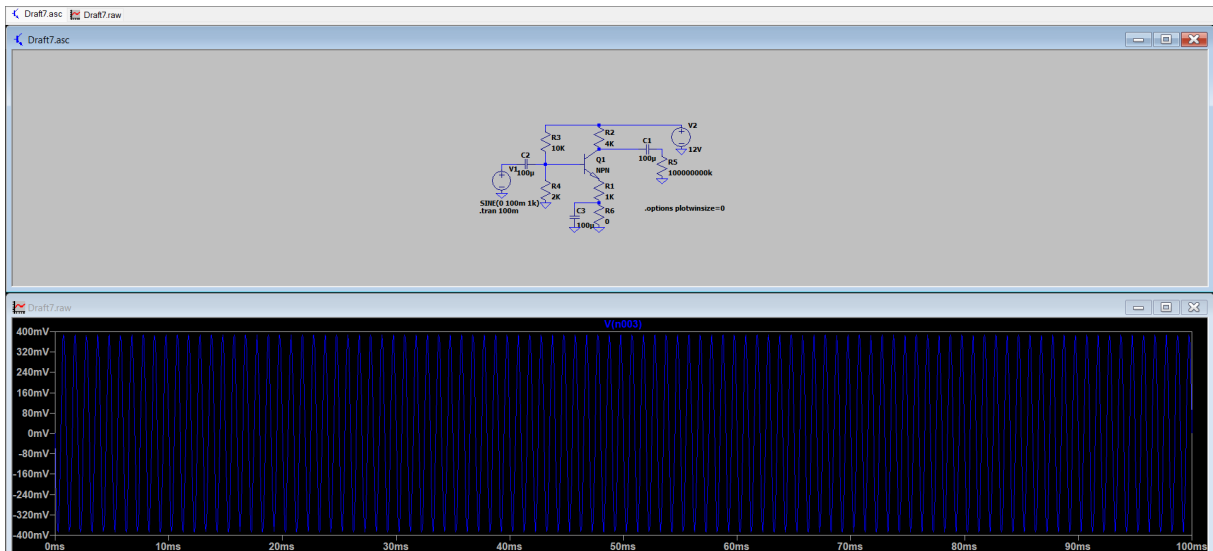
$$A_v = -\frac{R_C}{R_{E2}} \quad (43)$$

$R_{E2}$  can vary with  $R_{E1}$ , giving a variable gain. The bias is stable and the gain does not depend on process, voltage, or temperature.

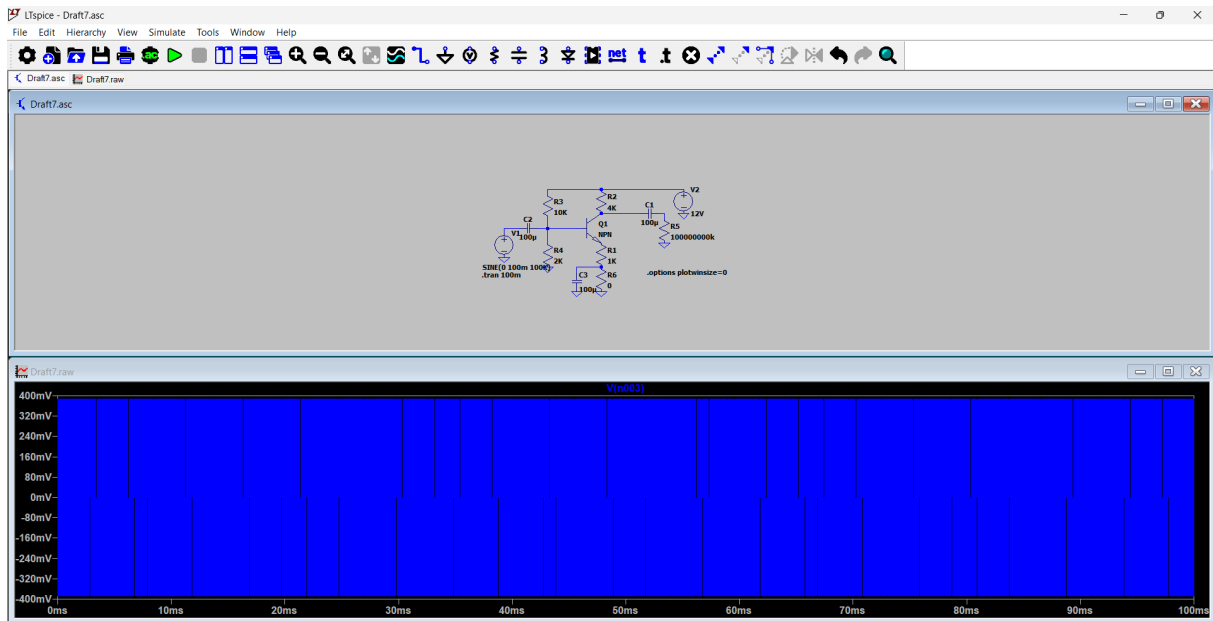
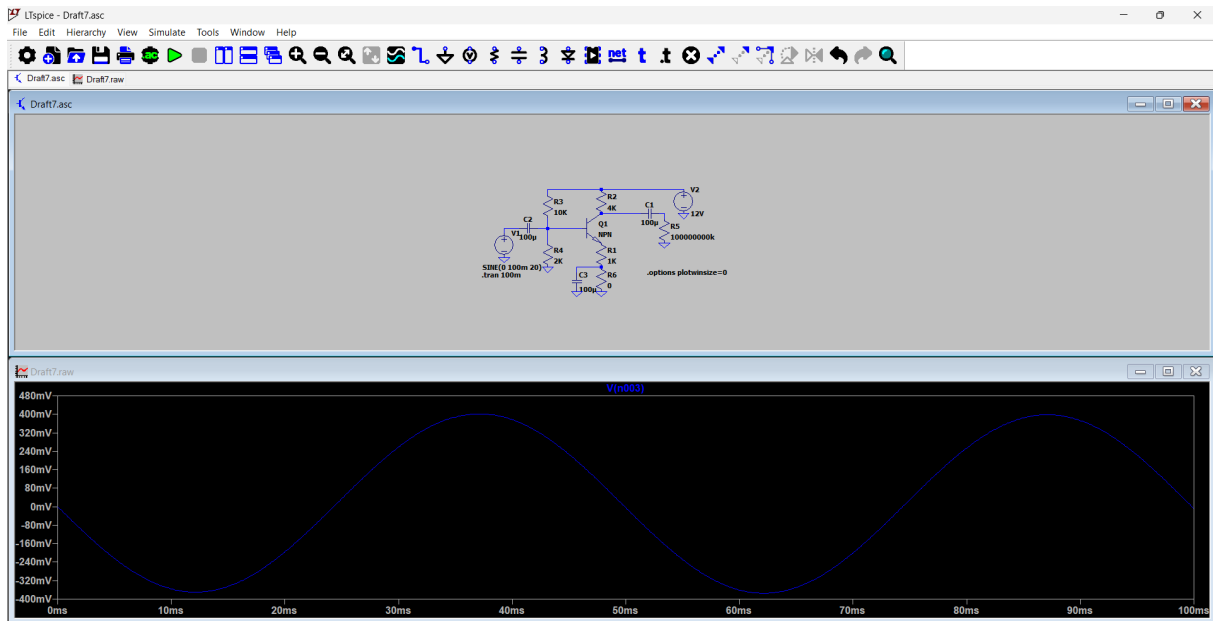
**Test range:** 20 Hz  $\rightarrow$  100 kHz, 1 mV  $\rightarrow$  50–100 mV

## Simulation

Now test the variable gain; convert it to 10.



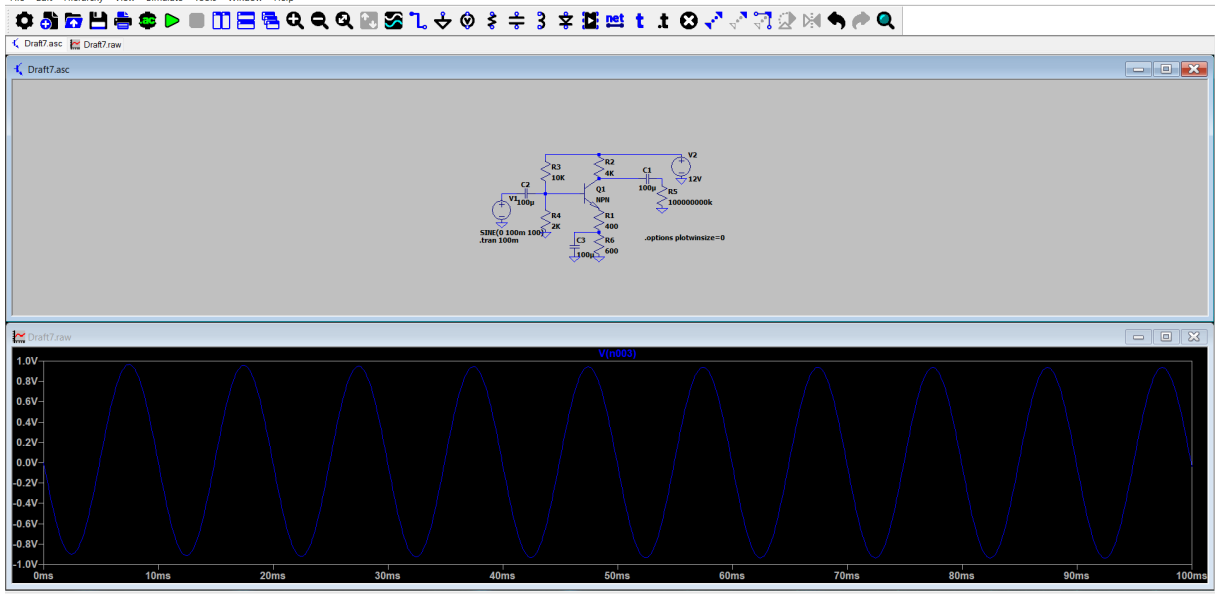




## Loading Effect Modification

$R_{\text{signal}} = 32 \Omega$ , which is small and this amplifier can work with it.

$R_{\text{load}} = 8 \Omega$ , which is very small, so we want a voltage buffer stage.



## Design

### First Stage:

$$R_o \simeq 4 \text{ k}\Omega$$

So the next stage input impedance:

$$R_{in} \simeq r_{\pi} + (\beta + 1)R_E \parallel R_1 \parallel R_2 \simeq R_1 \simeq 2 \text{ k}\Omega \quad (44)$$

$$V_{in2} = \frac{2}{2+4} V_{o1} \simeq \frac{V_o}{3} \quad (45)$$

### Buffer Stage (Emitter Follower):

$$A_v = \frac{R_E}{R_E + \alpha r_m}, \quad R_o = R_E \parallel \alpha r_m \quad (46)$$

$R_E$  must be large to make the gain  $\simeq 1$ , so  $R_E \simeq 1\text{--}5 \text{ k}\Omega$ .

$$r_m = \frac{V_T}{I_C} \simeq 26\text{--}50 \text{ }\Omega \quad (47)$$

$$R_o \simeq 42 \text{ }\Omega \quad (48)$$

Total gain:

$$A_{v,\text{total}} = A_{v1} \times \frac{1}{3} \times \frac{8}{8+42} = \frac{A_{v1}}{15} \quad (49)$$

Therefore:

$$A_{v1} \simeq 70\text{--}80 \quad (50)$$

$$R_E \simeq 50\text{--}60\ \Omega \simeq 60\ \Omega \quad (51)$$

**Simulation:**

